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Exercise 5 (14 points) – individual work

- The answers can be typed or handwritten (handwriting must be clear and readable), in this exercise sheet or your own sheet (put your name & ID at the top of the sheet). All answers must be saved to only 1 PDF file.
- Some questions also require the submission of processes/workflows (file.rmp or file.ipynb).
- In case of re-submission (after first grading) or submission after solution is given, your points will be weighted by 0.5.

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1. (Total 6 points)

Retrieve **cpu data**, which contains the following attributes. All of them are numeric.

Attribute	Description
MYCT	Machine cycle time (nanoseconds)
MMIN	Minimum main memory (kilobytes)
MMAX	Maximum main memory (kilobytes)
CACH	Cache memory (kilobytes)
CHMIN	Minimum channels (units)
CHMAX	Maximum channels (units)
PERF (target)	Performance score

1.1 (3 points) Predict PERF by linear regression method.

Instructions/Questions	Answers
Step 1. Run linear regression to predict PERF in split validation with split ratio = 0.7. When setting linear regression parameters, don't use feature selection & colinear feature elimination.	rmse = 59.523 rrse = 0.321 r = 0.952 r^2 = 0.907
Step 2. Run linear regression as in step 1. This time, use feature selection (pick 1 method that gives best result) & colinear feature elimination.	I use forward selection to choose select attribute rmse = 52.664 rrse = 0.521 r = 0.899 r^2 = 0.808
Step 3. Compare results from steps 1 and 2. Identify attribute(s) that can be considered redundant or can be excluded from the regression model.	Redundant attributes = MYCT, CACH, CHMIN are eliminated if use forward selection
Also submit your workflow that performs both steps. The output in each step must include linear regression table and	

performance vector (RMSE, RRSE, r, and r^2). Name the workflow **question1_1.rmp**.	

1.2 (3 points) Use 1 machine learning method (decision tree, SVR, neural network) to predict PERF instead. Try to get better performance than linear regression. If any data preprocessing is needed, do it.

Instructions/Questions	Answers
Submit your workflow. Name the workflow **question1_2.rmp**. Make sure that it output the same performance vector as in 1.1.	Your method = Neural network
Briefly explain which data preprocessing is applied to which attribute (if there is any).	I apply normalization because data is on different scales, some in thousands, some are just in tens. (Does not use a separate block, just check normalized neural network.
Briefly explain how the method's parameters are set differently from classification task.	Classification is trying to predict what class the records, regression predicts numerical value. As for parameters for regression and classification, obviously different models will use different hyper parameters since they are based on different math and principles. They also use different evaluation metrics regression use rmse, rrse (some from of error value from the real value). Classification use mainly cross entropy as error.

Name: Pakin Panawattanakul ID: 6580043

2. (Total 8 points) Use training data in the following table to answer questions

2.1 (3 points) Consider AdaBoost model trained by the following training data.

Training Data				
Record	A	B	C	Buy (class)
1	T	T	positive	yes
2	F	T	negative	no
3	F	F	neutral	yes
4	T	F	negative	no
5	F	T	positive	no
6	T	T	positive	yes
7	F	T	negative	no
8	T	F	positive	yes
9	F	F	negative	no
10	T	T	neutral	yes

AdaBoost model
Base classifier 1, weight = 2.197 if C = negative then buy = no if C = neutral then buy = yes if C = positive then buy = yes
Base classifier 2, weight = 2.079 if A = F && B = F then buy = yes if A = F && B = T then buy = no if A = T && B = F then buy = no if A = T && B = T then buy = yes
Base classifier 3, weight = 2.708 if A = F then buy = no if A = T then buy = yes

(a) During 3 rounds of training base classifiers, which training records are easiest to predict?

Solutions are written in: Round = (pred, pred == expected)

Base classifier 1:

R1 = (yes, True)
R2 = (no, True)
R3 = (yes, True)
R4 = (no, True)
R5 = (yes, False)
R6 = (yes, True)
R7 = (no, True)
R8 = (yes, True)
R9 = (no, True)
R10 = (yes, True)

Base classifier 2:

R1 = (yes, True)
R2 = (no, True)
R3 = (yes, True)
R4 = (no, True)
R5 = (no, False)
R6 = (yes, True)
R7 = (no, True)
R8 = (no, False)
R9 = (yes, False)
R10 = (yes, True)

Base classifier 3

 Name: Pakin Panawattanakul ID: 6580043

R1 = (yes, True) R2 = (no, True) R3 = (yes, True) R4 = (yes, False) R5 = (yes, False) R6 = (yes, True) R7 = (no, True) R8 = (Yes, True) R9 = (yes, False) R10 = (yes, True) R [5, 8, 9, 3, 4] each have one wrong prediction Easy to predict = R[1, 2, 6 ,7, 10]
(b) Find ensemble predictions for record 3 and for record 5. R3 = 2.197 + 2.079 - 2.708 = 1.568 : pred = y R5 = 2.197 - 2.079 - 2.708 = 2.590 : pred = n

Name: Pakin Panawattanakul ID: 6580043

2.2 (2 points) After training 5 weak classifiers, their predictions on training data are as follows.

Row No.	record	buy	base_prediction0	base_prediction1	base_prediction2	base_prediction3	base_prediction4
1	1	yes	yes	yes	yes	yes	no
2	2	no	no	no	no	no	no
3	3	yes	no	no	yes	no	no
4	4	no	no	yes	yes	yes	yes
5	5	no	no	yes	no	no	yes
6	6	yes	yes	yes	yes	yes	no
7	7	no	no	no	no	no	no
8	8	yes	yes	yes	yes	no	yes
9	9	no	no	no	yes	yes	no
10	10	yes	yes	yes	yes	yes	yes

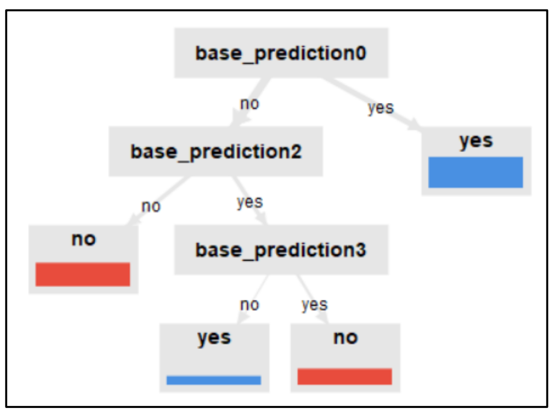
(a) Using voting ensemble by majority vote, find ensemble prediction for record 3.

Pred = no

(b) Find the training accuracy of this voting model (when it predicts all training records).

Solutions are written in: Round = (pred, pred == expected)
R1 = (yes, True)
R2 = (no, True)
R3 = (no, False)
R4 = (yes, False)
R5 = (no, True)
R6 = (yes, True)
R7 = (no, True)
R8 = (yes, True)
R9 = (no, True)
R10 = (yes, True)

2.3 (3 points) Suppose that we train meta decision tree classifier to learn the prediction patterns of 5 weak classifiers in 2.2 (i.e. the table in 2.2 is training partition for meta training). The meta model is as follows.



 Name: Pakin Panawattanakul ID: 6580043

(a) Find ensemble prediction for record 3. Pred = yes
(b) From the meta model, list all base classifiers that contribute to the final prediction. Base classifier 0, 2, 3
(c) From the meta model, list all base classifiers that don't contribute to the final prediction Base classifier 1,4